

AgenticCyOps

Baseline Verification Report

Domain: Healthcare

May 24, 2026 21:34

Configurations tested: Flat MAS | ACL-Hardened | AgenticCyOps

Securing Multi-Agent AI Integration in Enterprise Cyber Operations

Experiment Configuration

Model	Role	Family	GPU Assignment	Port
Qwen3-235B-A22B-Instruct-2	Primary Agents + Host	Qwen	GPU 0,1,4,5 (TP=4)	8000
	Validator V1	Qwen	GPU 2	8002
DeepSeek-R1-Distill-Qwen-3	Validator V2	DeepSeek	GPU 3	8005
	Claude Sonnet	Anthropic	API	--
	GPT-4o	OpenAI	API	--
	Qwen3-Embedding-0.6B	Qwen	CPU	--

Model & GPU Assignment

Three-Configuration Comparison

Behavior	Flat MAS	ACL-Hardened	AgenticCyOps
Tool visibility	All 16 tools	All 16 tools	Manifest only (4-5)
Tool enforcement	None	ACL (HTTP 403)	P2 Manifest Enforcer
Consensus (P3)	Disabled	Disabled	V1+V2+V4+V6 (3/4)
Memory access	Direct	Direct + ACL	MMA Gateway (P4+P5)
Write filtering (P4)	None	None	Cosine sim > 0.5
Escalation	Never	Never	On bulk/rejection

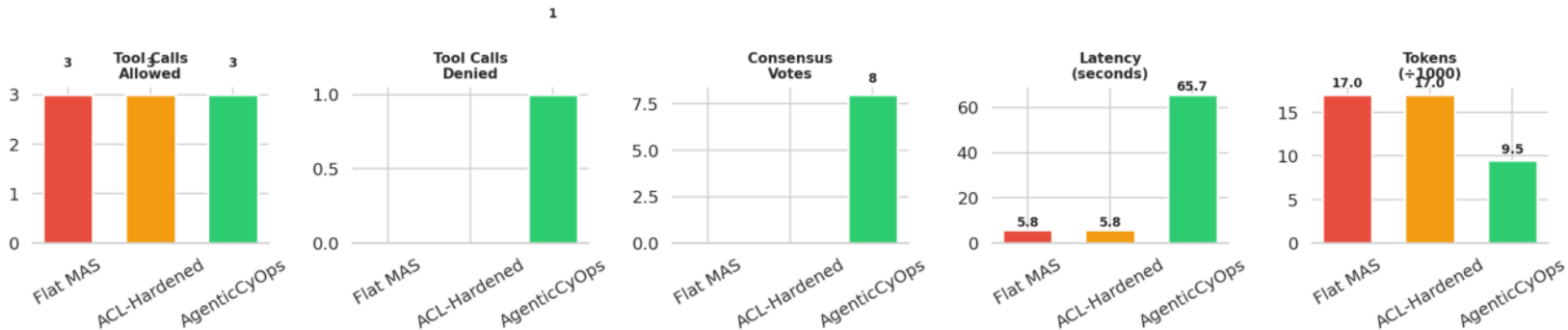
Baseline Metrics — Healthcare

Metric	Flat MAS	ACL-Hardened	AgenticCyOps
Total Events	15	15	54
Tool Calls Attempted	3	3	4
Tool Calls Allowed	3	3	3
Tool Calls Denied	0	0	1
Consensus Votes	0	0	8
E2E Latency (s)	5.8	5.8	65.7
Total Tokens	17,039	17,039	9,475
Phases Completed	4	4	4

AgenticCyOps uses 44% fewer tokens than Flat MAS

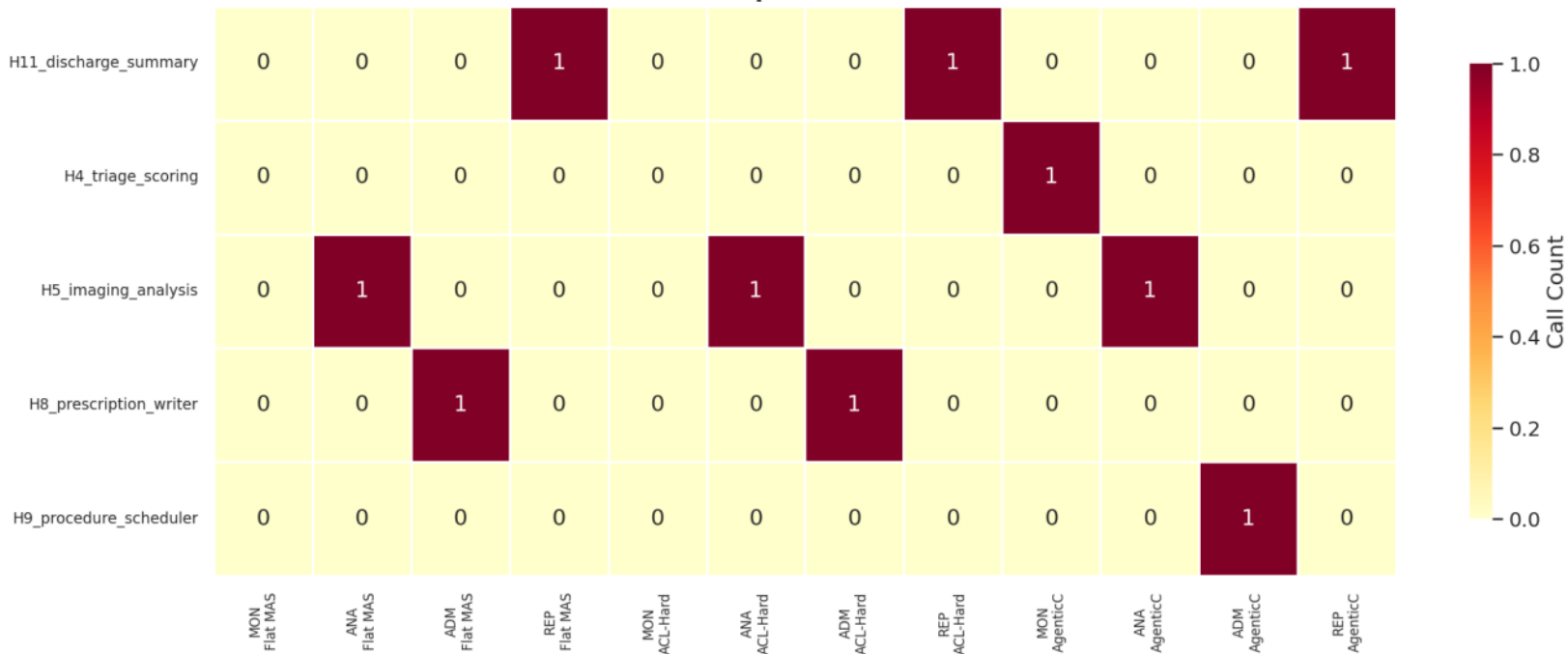
Configuration Comparison

Configuration Comparison — Healthcare Baseline

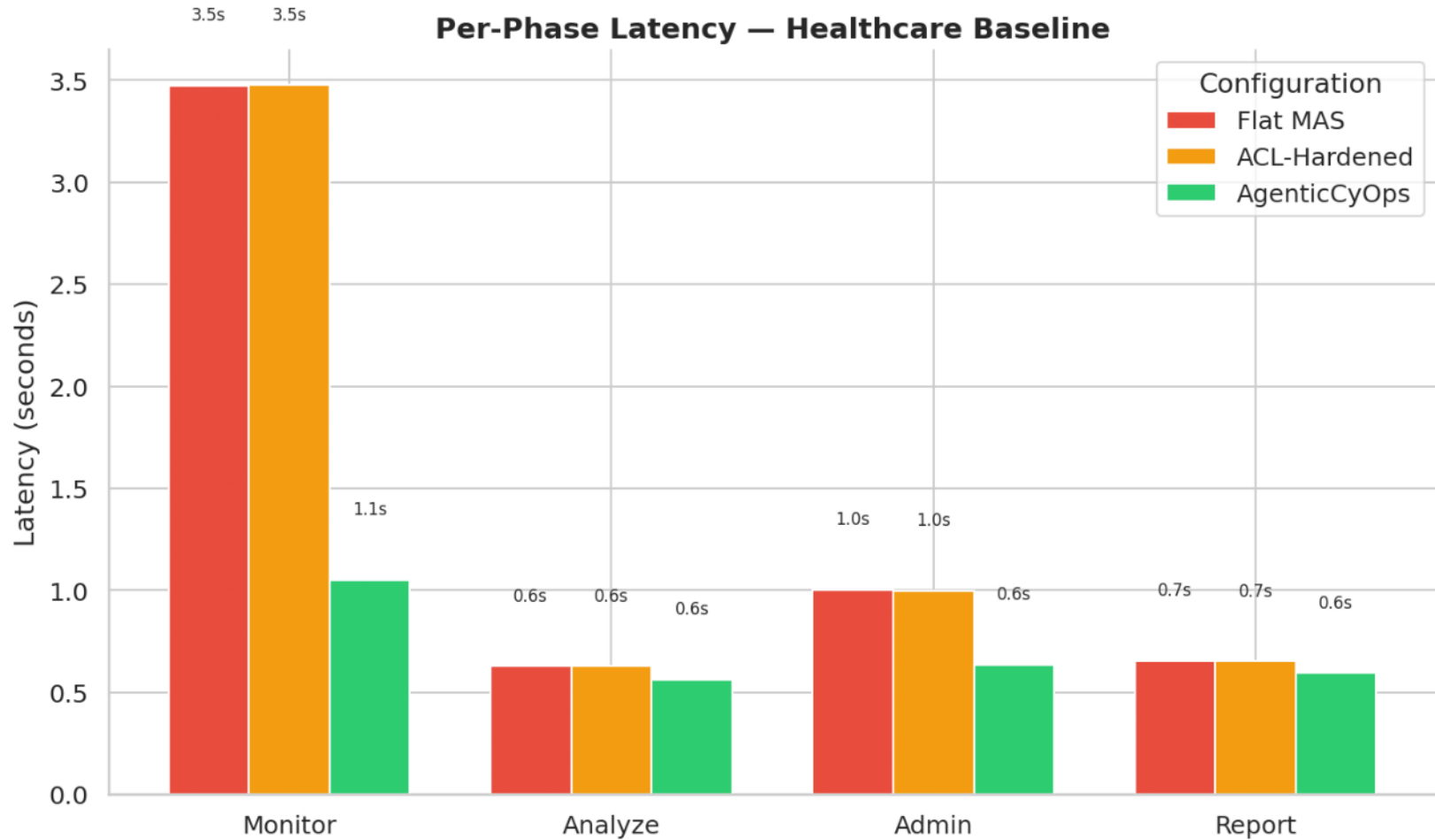


Tool Call Distribution

Tool Call Heatmap — Healthcare Baseline

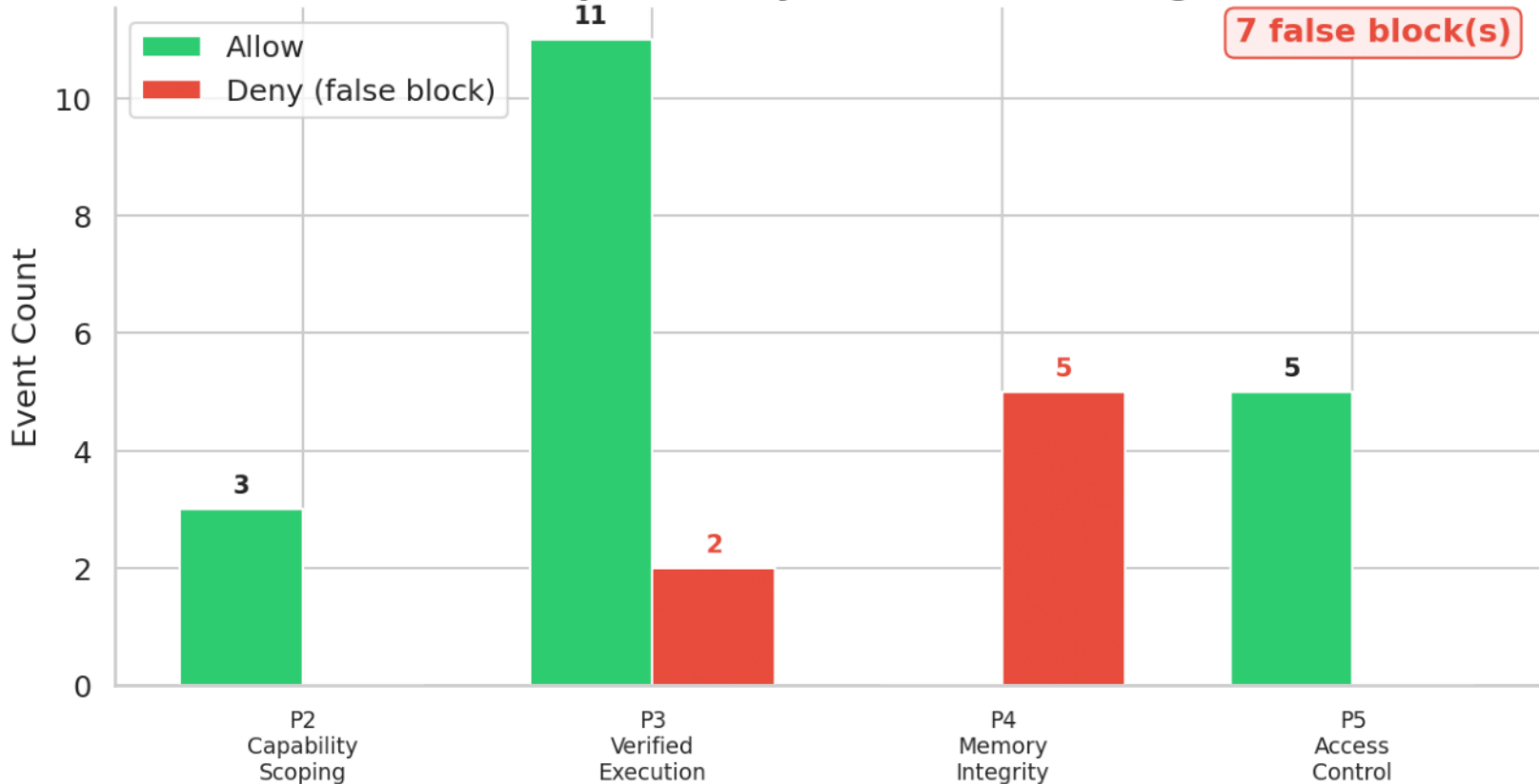


Per-Phase Latency



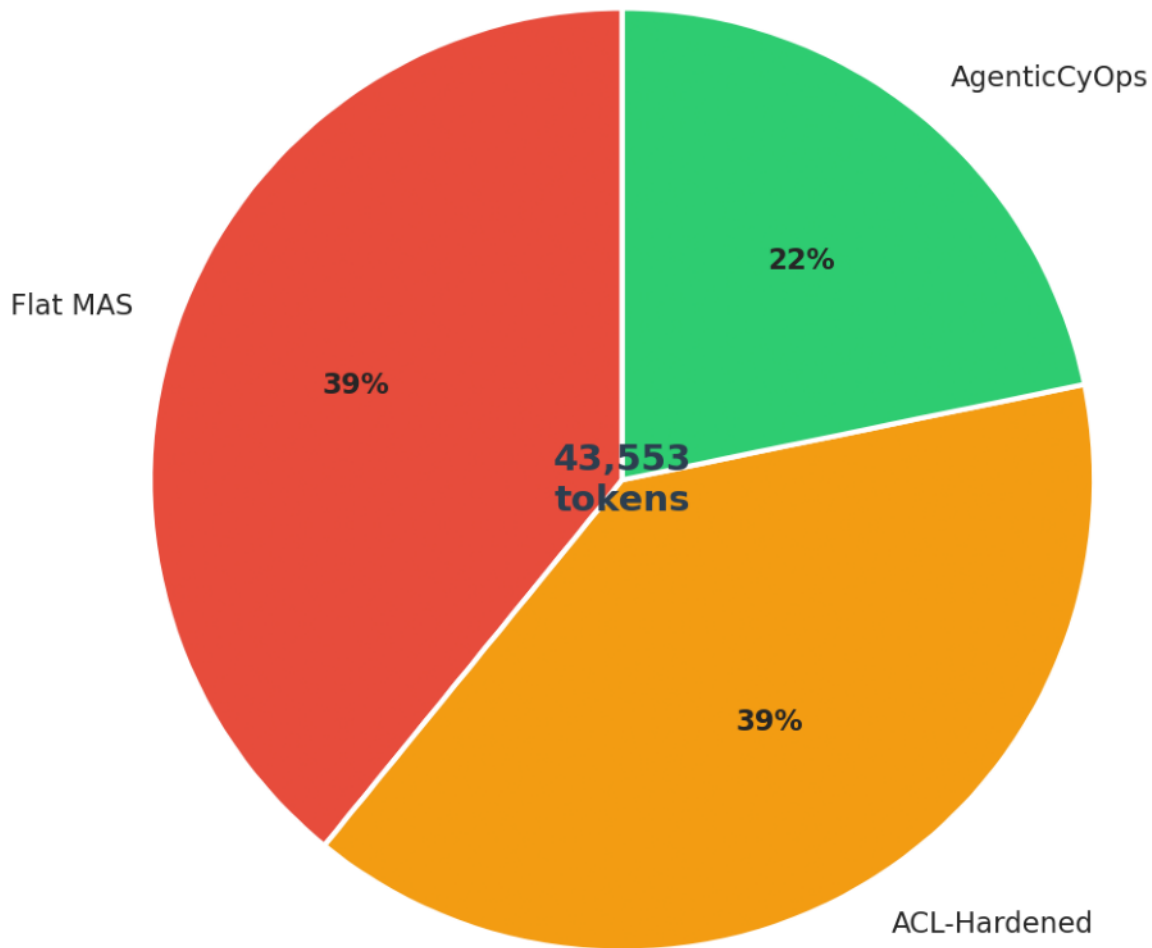
Defensive Principle Activity

Defensive Principle Activity — Healthcare Benign Baseline



Token Distribution

Token Distribution — Healthcare Baseline



Key Findings

- + Token reduction: AgenticCyOps uses 44% fewer tokens than Flat MAS
- + Consensus active: 8 validator votes recorded
- False blocks: 1 benign calls denied by P3 consensus
- + Pipeline complete: all 4 phases executed successfully
- ACL enforcement: 0 out-of-scope calls blocked (HTTP 403)
- Flat baseline: 3 tool calls with zero enforcement

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Domain: Healthcare | Configs: 3 | Models: 5 families